

GitHub for Collaboration

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with special thanks to
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Get a GitHub Account

- MINKWIC_2025 (Wifi & password)
- Go to <https://github.com>
- Click SignUp
 - Email
 - Password
 - Username (This is visible on the web)
- Go to <https://github.com/cigas>
- Scroll down to MINKWIC2025



Reasons for you to use GitHub

- Portfolio
- Version control
- Team coding



GitHub Details

- *Centralized* service - hosts your repositories
- Uses Git (inherently distributed)
- Provides many other coding/development tools



Portfolio

- Show off your projects
- Share your work with the community
- Use markdown for simple display
- GitHub pages (<https://docs.github.com/en/pages>)
for nicely built complete web site



Building a Portfolio

- Fork an existing one
 - Easy to get started
 - But it's not your work...
- Create your own
 - One repository for each distinct project
 - Use folders!
 - Add your own code
 - Edit [README.md](#) for each repository



Fork

- Get a copy of another repository to modify
 - Find useful code
 - Contribute to open source project
- Fork my workshop demo repository





MINKWIC2025

Public



Pin



Watch

0



Fork

0



Star

0

Fork your own copy of cigas/MINKWIC2025

Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project.

Required fields are marked with an asterisk ().*

Owner *

Choose an owner

/

Repository name *

MINKWIC2025

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

A demo repository for MINKWIC 2025 workshop

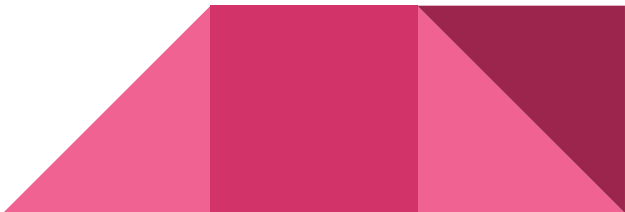
43 / 350 characters

☒ Copy the `main` branch only

Contribute back to cigas/MINKWIC2025 by adding your own branch. [Learn more.](#)

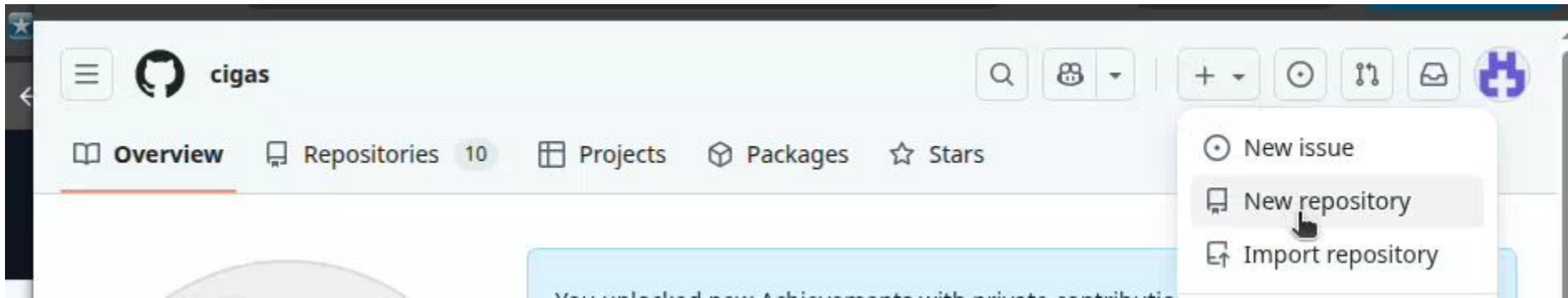
Create fork

Demo Repository

- README describes contents
 - AoC contains Advent of Code solutions
 - To Do:
 - Edit [README.md](#)
 - Add your name
 - Commit
- 

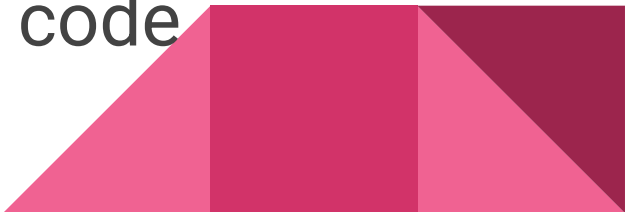
Create your own repository

- Homepage, +, New repository



- Include README and license

Portfolio Contents

- Things to include:
 - Independent study projects
 - Code from contests/challenges
 - Document your work
 - Potential employers see your work
 - Helps others who can use your code
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Portfolio Contents

- Class assignment solutions can cause problems
 - Some schools have policies against this
 - Ethics considerations of posting solutions
 - Is an assignment solution really your best work to show off?



Version Control

- Code stored on github.com
 - Off-site backup for free
 - Every commit is saved forever
 - Efficient storage - only save actual changes
- Revert back to previous version
 - Add another copy of previous version as current one



Version Control (cont.)

- Branches
 - Change algorithm - compare results
 - Production vs. development
 - Solution vs. starter code (careful)
- Document sets of changes, but not in code itself
 - Good commit messages



Normal Workflow

- Edit README on GitHub - OK
- Edit code on your local computer
 - Use IDE with Git integration
 - Check out a copy of repository
 - Within IDE
 - Commit to your local repository
 - Push back to GitHub



Team Collaboration

- Multiple concurrent editors
 - Automatically handle changes w/o conflicts
 - Flag conflicts for manual resolution
 - Pull requests for hierarchical collaboration
 - Work on a fork
 - Submit a PR with your documented changes
 - Moderator reviews - accepts and merges
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Contact

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<https://github.com/cigas/MINKWIC2025>